7+1: the octonions and the Clifford algebra of 7+1 spacetime

A real realization of the octonions, and a real 16×16 realization of the Clifford algebra of 7+1 spacetime, built from — and mirroring — the split octonions and the 16×16 Dirac matrices of the notebook 4+4.nb

Repository: github.com/once-ere/7+1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1), evaluated Mon 26 Oct 2026 14:54:43
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Abstract

Starting from the real 8×8 matrices τ_A and the real 16×16 Dirac matrices T_A of 4+4.nb (split octonions and the Clifford algebra $Cl(4,4)$), this document and the notebook 7+1.nb construct (i) a real realization of the octonions whose multiplication constants come from exactly the same vielbein formula as the split-octonion constants of 4+4.nb; (ii) real 16×16 generators of the Clifford algebra of 7+1 spacetime (one time, seven space dimensions), built only from the matrices of 4+4.nb; (iii) a proof that in the sign convention of 4+4.nb no real 16×16 realization exists, together with the minimal complexification and a real 32×32 realization; (iv) examples that test everything. All 128 checks of the notebook are True.

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1. Results at a glance

Octonions from the split-octonion matrices. With $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ (the 8×8 matrix $\mathfrak{o}$ of 4+4.nb),

$$\tau_0^O = 1_8, \qquad \tau_A^O = \tau_A \;\; (A=1,2,3), \qquad \tau_A^O = \tau_A\,\sigma \;\; (A=4,5,6,7), \qquad \bar\tau_A^O = (\tau_A^O)^{\mathsf T}.$$

These are real, $\tau_A^O\bar\tau_B^O + \tau_B^O\bar\tau_A^O = 2\,\delta_{AB}\,1_8$, and the vielbein formula of 4+4.nb turns them into the multiplication table of the octonions $\mathbb{O}$ (section 6).

Real Clifford algebra of 7+1 spacetime. With the 16×16 Dirac matrices T_A of 4+4.nb and $S_{44} = T_4T_5T_6T_7$,

$$\Gamma_0^{(1,7)} = T_0, \qquad \Gamma_i^{(1,7)} = T_0\,T_i \;\; (i=1,2,3), \qquad \Gamma_k^{(1,7)} = T_0\,T_k\,S_{44} \;\; (k=4,5,6,7),$$

$$\Gamma_A^{(1,7)}\Gamma_B^{(1,7)} + \Gamma_B^{(1,7)}\Gamma_A^{(1,7)} = 2\,\eta_{AB}^{(1,7)}\,1_{16}, \qquad \eta^{(1,7)} = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1).$$

All entries are $0,\pm1$. Time (x_0, the octonion unit) squares to $+1$, the seven space directions to -1. The quadratic form is the real part of the octonion square, $\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2$; the algebra is $Cl(1,7) \cong \mathbb{R}(16)$ (section 7.2).

The sign convention of 4+4.nb. 4+4.nb writes metrics as "space-like = +1, time-like = −1". In that convention the 7+1 metric is $\eta^{(7,1)} = \mathrm{diag}(-1,+1,\ldots,+1) = -\eta^{(1,7)}$ and the relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}^{(7,1)}1_{16}$ defines $Cl(7,1) \cong \mathbb{H}(8)$, which has NO real 16×16 realization, whatever the order or signs of the indices (theorem of section 7.4). The minimal complexification multiplies only the time block by i and keeps the exact block form of 4+4.nb:

$$\Gamma_A^{(7,1)} = \begin{pmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{pmatrix}, \qquad \tau_0^C = \bar\tau_0^C = i\,1_8, \quad \tau_k^C = \tau_k^O,\; \bar\tau_k^C = \bar\tau_k^O \;\; (k=1,\ldots,7),$$

$$\Gamma_A^{(7,1)}\Gamma_B^{(7,1)} + \Gamma_B^{(7,1)}\Gamma_A^{(7,1)} = 2\,\eta_{AB}^{(7,1)}\,1_{16}.$$

A real realization of $Cl(7,1)$ needs 32×32 matrices: $\widehat\Gamma_A^{(7,1)} = \varepsilon \otimes \Gamma_A^{(1,7)}$ with $\varepsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (section 7.5).

Verification. The notebook 7+1.nb contains 128 checks; evaluated in 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026) it returns True for every one of them, with 0 message cells, in 5.09545 s. An independent Python program reproduces every matrix and table that the notebook prints (section 9).

2. Coordinates, metrics and conventions

4+4.nb and 7+1.nb use the same eight coordinate symbols $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (Mathematica: `X = {x0, x1, x2, x3, x4, x5, x6, x7}`; never $x_1, \ldots, x_8$). Their meaning differs:

coordinate	meaning in 4+4.nb (signature 4+4)	meaning in 7+1.nb (signature 7+1)
x_0	hidden space-like dimension	ordinary time; octonion unit $e_0 = 1$
x_1, x_2, x_3	ordinary 3-space	ordinary 3-space; imaginary units e_1, e_2, e_3
x_4	ordinary time	auxiliary space-like; $e_4 = \ell$ (Cayley–Dickson doubling unit)

coordinate	meaning in 4+4.nb (signature 4+4)	meaning in 7+1.nb (signature 7+1)
x_5, x_6, x_7	three auxiliary time-like dimensions ("superluminal deflating time")	auxiliary space-like; e_5, e_6, e_7

Metrics (Mathematica names in brackets):

$\eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$	[η4488] (4+4.nb: space-like $+1$, time-like -1)
$\eta^{(7,1)} = \mathrm{diag}(-1,+1,+1,+1,+1,+1,+1,+1)$	[η71] (7+1 in the sign convention of 4+4.nb)
$\eta^{(1,7)} = -\eta^{(7,1)} = \mathrm{diag}(+1,-1,\ldots,-1)$	[η17] (7+1, octonionic convention)
$g^O = 1_8$	[g0] (norm of the octonions)

Conventions. Indices $A, B, K, L \in \{0,\ldots,7\}$ label coordinates, octonion units and Clifford generators; $i, j \in \{1,\ldots,7\}$ label imaginary units and space directions; $h \in \{1,2,3\}$. 1_n is the identity matrix, M^{T} the transpose, δ_{AB} the Kronecker delta, $\{X,Y\} = XY + YX$, and $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ a matrix of blocks. The index 8 on a Clifford generator denotes the ordered product of all eight, as in 4+4.nb ($T_8 = T_0 T_1 \cdots T_7$). $Cl(p,q)$ always means: p generators square to $+1$ and q to -1, i.e. $v^2 = Q(v)$ with Q of signature (p,q). Mathematica lists are 1-based, so the metric entry η_{AB} is written `η[[A+1, B+1]]` in the notebook.

3. Notation used in this document

Every symbol of this document, with its Mathematica name in 7+1.nb and its definition. Names in the first group are reproduced exactly from 4+4.nb (where some of them carried 2-D names made with the Notation package: $\bar\tau_A$ was a symbolized tau-bar, T_A was `T16^A[A]`, E_{44} was `E^A_a[h]`); the plain one-line names avoid confusing them with powers or indices.

symbol	Mathematica	definition / meaning
σ	`σ`	$\left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$, split metric on spinor space (4+4.nb `σ88`)
$s_h,\ t_h$	`s4by4[h]`, `t4by4[h]`	self-dual / anti-self-dual 4×4 matrices, $(s_h)_{pq} = \epsilon_{hpq4} - (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$, $(t_h)_{pq} = \epsilon_{hpq4} + (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$
ζ_A	`ζ[A]`	$\zeta_0 = 1_8$, $\zeta_h = \left(\begin{smallmatrix} 0 & s_h \\ s_h & 0 \end{smallmatrix}\right)$, $\zeta_{7-h} = \left(\begin{smallmatrix} 0 & t_h \\ -t_h & 0 \end{smallmatrix}\right)$, $\zeta_7 = \zeta_1 \cdots \zeta_6$
$\tau_A,\ \bar\tau_A$	`τ[A]`, `τbar[A]`	$\tau_A = \zeta_A$, $\bar\tau_A = \sigma \tau_A^{\mathsf{T}} \sigma$ (split octonions)
T_A	`T16[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, Dirac matrices of 4+4.nb; $T_8 = T_0 \cdots T_7$
$P_L,\ P_R$	`PL44`, `PR44`	$(1_{16} \mp T_8)/2$
u	`unit`	$(1,0,0,0,1,0,0,0)/\sqrt{2}$, the unit spinor (`uBasis[[8]]`, `hUSE = 8`)
E_{44}	`E44[8]`	split vielbein, rows $u^{\mathsf{T}} \sigma \bar\tau_B$
ε_B	`eA[B]`	split-octonion units, $B = 1,\ldots,8$
$\tau^O_A,\ \bar\tau^O_A$	`τO[A]`, `τObar[A]`	octonion matrices, section 6.1
E^O	`EO`	octonion vielbein, rows $u^{\mathsf{T}} \bar\tau^O_B$
L^O_A	`LO[A]`	$E^O \tau^O_A (E^O)^{\mathsf{T}}$, left multiplication by e_A
e_A	`e[A]`	octonion units, $A = 0,\ldots,7$, $e_0 = 1$
$N(x),\ \bar x,\ \mathrm{Re}\, x$	`oNorm2`, `oConj`, `oRe`	norm, conjugate, real part
T^O_A	`T16O[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau^O_A \\ \tau^O_A & 0 \end{smallmatrix}\right)$, generators of $Cl(8,0)$
S_{44}	`S44`	$T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma,\sigma)$
$\Gamma^{(1,7)}_A$	`Γ17[A]`	real generators of the 7+1 Clifford algebra, metric $\eta^{(1,7)}$
$\Gamma^{(7,1)}_A$	`Γ71[A]`	complex generators, metric $\eta^{(7,1)}$
$\widehat\Gamma^{(7,1)}_A$	`Γ71R32[A]`	real 32×32 generators, metric $\eta^{(7,1)}$
$\Gamma^{(5,3)}_A,\ \Gamma^{(0,8)}_A$	`Γ53[A]`, `Γ08[A]`	real generators of $Cl(5,3)$ and $Cl(0,8)$ (section 8.6)
Σ_{AB}	`Σ17[A,B]`	$(\Gamma^{(1,7)}_A \Gamma^{(1,7)}_B - \Gamma^{(1,7)}_B \Gamma^{(1,7)}_A)/4$
ε	`ε2`	$\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$
$\sigma_1,\ \sigma_3$	—	$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$
$\Gamma^{(1,7)}_8$	`Γ17[8]`	$\Gamma^{(1,7)}_0 \Gamma^{(1,7)}_1 \cdots \Gamma^{(1,7)}_7 = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, the volume element
$Q_i,\ Q_j,\ Q_k$	`Qi`, `Qj`, `Qk`	$\varepsilon \otimes 1_{16}$, $\sigma_1 \otimes \Gamma^{(1,7)}_8$, $\sigma_3 \otimes \Gamma^{(1,7)}_8$ (quaternion units, section 7.5)
$\chi^{(7,1)},\ \chi^{(1,7)}$	`χ71`, `χ17`	$-i\,\Gamma^{(7,1)}_8$ and $i\,\Gamma^{(1,7)}_8$, chirality matrices (section 7.6)

symbol	Mathematica	definition / meaning
$\vec{x}$	`xImag`	imaginary part $x_1e_1+\cdots+x_7e_7$ of the octonion x (section 7.7)
$S(\beta),\ R(\theta),\ \Lambda$	`Sβ`, `Rθ`, `Λboost`, `Λrot`	a boost, a rotation and their Lorentz matrices (section 8.2)
Γ^A	`Γ17up[A]`	$\eta^{AA}_{(1,7)}\Gamma^{(1,7)}_A$, index raised with the inverse metric
$p,\ m,\ \not{p}$	`pvec`, `mass`, `pslash`	$p=(3,2,2,0,0,0,0,0)$, $m=1$, $\not{p}=\sum_A p^A\Gamma^{(1,7)}_A$ (section 8.4)
W	`realAmplitudes`	the 8 real solutions w of $(\not{p}-m)w=0$, one per row
$k,\ \not{k}$	`kvec`, `kslash`	$k=(1,1,0,0,0,0,0,0)$, $\not{k}=\sum_A k^A\Gamma^{(1,7)}_A$ (section 8.5)
$\widehat{\Gamma}^A\partial_A,\ P$	`DiracOp71R32`, `pslash32`	$\sum_A \eta^{AA}_{(7,1)}\widehat{\Gamma}^{(7,1)}_A\partial_A$ and $P=\sum_A p^A\widehat{\Gamma}^{(7,1)}_A$ (section 8.5)

4. The two definitions, and four corrections

4.1 [1A] The octonions

The octonions $\mathbb{O}$ form an 8-dimensional, non-associative, non-commutative normed division algebra over $\mathbb{R}$, with basis $e_0=1,e_1,\ldots,e_7$:

$$\begin{aligned}
x&=\textstyle\sum_{A=0}^{7}x_Ae_A, \qquad e_i^2=-1\ (i=1,\ldots,7), \qquad e_ie_j=-e_je_i\ (i\neq j),\\
\bar{x}&=x_0-\textstyle\sum_{i=1}^{7}x_ie_i, \qquad \overline{xy}=\bar{y}\,\bar{x}, \qquad \mathrm{Re}(x)=\tfrac{1}{2}(x+\bar{x})=x_0,\\
N(x)&=x\bar{x}=\bar{x}x=x_0^2+x_1^2+\cdots+x_7^2, \qquad N(xy)=N(x)N(y), \qquad x^{-1}=\bar{x}/N(x),\\
x(xy)&=(xx)y, \qquad (yx)x=y(xx) \qquad \text{(alternativity; power-associativity follows).}
\end{aligned}$$

Products of distinct imaginary units follow the oriented lines of the Fano plane, each line spanning a quaternion subalgebra. Cayley–Dickson: $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$ as pairs of quaternions.

4.2 [1B] The Clifford algebra of an 8-dimensional space

$$\begin{aligned}
Cl(V,Q)&=\mathcal{T}(V)/\mathcal{I}_Q(V), \qquad \mathcal{I}_Q(V)=\big\langle v\otimes v-Q(v)1\big\rangle,\\
v^2&=Q(v)\,1, \qquad vw+wv=2B(v,w)\,1, \qquad B(v,w)=\tfrac{1}{2}\big(Q(v+w)-Q(v)-Q(w)\big),\\
e_A^2&=Q(e_A), \qquad e_Ae_B=-e_Be_A\ (A\neq B), \qquad \dim Cl(V,Q)=2^8=\textstyle\sum_{k=0}^{8}\binom{8}{k}=256,\\
Cl_8(\mathbb{C})&\cong\mathbb{C}(16).
\end{aligned}$$

4.3 Corrections (each verified in 7+1.nb)

1. **Cayley–Dickson formula.** The second formula of [1A], $(a,b)(c,d)=(ac-\bar{d}b,\ \bar{a}d+cb)$, does not define the octonions: it is not alternative and its norm is not multiplicative. Counterexample: $x=(1,1)=e_0+e_4$, $y=(i,0)=e_1$ give $x^2=(0,2)$, $x^2y=(0,2i)$, but $xy=(i,i)$ and $x(xy)=(2i,2i)$. A correct formula of the same shape is $(a,b)(c,d)=(ac-d\bar{b},\ \bar{a}d+cb)$ (J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002), section 2.2); the octonions of this document satisfy it exactly with $x=a+\ell b$, $\ell=e_4$. The first formula of [1A] contains a stray symbol "x" and cannot be read unambiguously.

2. **Grade 4.** $\binom{8}{4}=70$, not 72: $1+8+28+56+70+56+28+8+1=256$.

3. **Classification.** With $v^2=Q(v)$ and $p=$ number of generators squaring to $+1$, the correct table is the one below. $\mathbb{C}(16)$ can never occur: its real dimension is $2\cdot16^2=512$, while every $Cl(p,q)$ with $p+q=8$ has real dimension $2^8=256$. Several "Metric Signs" entries of [1B] were also mistyped.

4. **Split octonions are not a division algebra.** In the split-octonion table of 4+4.nb, $(\varepsilon_1+\varepsilon_5)(\varepsilon_1-\varepsilon_5)=\varepsilon_1-\varepsilon_5^2=0$ although both factors are nonzero. They form a composition algebra with an indefinite norm of signature (4,4).

$Cl(p,q)$	**squares of the generators**	**correct**	**claimed in [1B]**	**status**
$Cl(8,0)$	+,+,+,+,+,+,+,+	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees
$Cl(7,1)$	+,+,+,+,+,+,+,−	$\mathbb{H}(8)$	$\mathbb{R}(16)$	wrong in [1B]
$Cl(6,2)$	+,+,+,+,+,+,−,−	$\mathbb{H}(8)$	$\mathbb{H}(8)$	agrees (signs mistyped)
$Cl(5,3)$	+,+,+,+,+,−,−,−	$\mathbb{R}(16)$	$\mathbb{C}(16)$	wrong in [1B]
$Cl(4,4)$	+,+,+,+,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees (signs mistyped)
$Cl(3,5)$	+,+,+,−,−,−,−,−	$\mathbb{H}(8)$	$\mathbb{C}(16)$	wrong in [1B]
$Cl(2,6)$	+,+,−,−,−,−,−,−	$\mathbb{H}(8)$	$\mathbb{H}(8)$	agrees (signs mistyped)
$Cl(1,7)$	+,−,−,−,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees
$Cl(0,8)$	−,−,−,−,−,−,−,−	$\mathbb{R}(16)$	$\mathbb{R}(16)$	agrees

Consequence for this project. With $v^2 = Q(v)$ and the sign convention of 4+4.nb ($Q = \eta^{(7,1)}$) the Clifford algebra of 7+1 spacetime is $Cl(7,1) \cong \mathbb{H}(8)$: no real 16×16 matrices exist (section 7.4). With the opposite overall sign ($Q = \eta^{(1,7)}$, the octonions' own form $\mathrm{Re}(xx)$) it is $Cl(1,7) \cong \mathbb{R}(16)$, realized by the real matrices $\Gamma_A^{(1,7)}$ built from 4+4.nb (section 7.2). Both describe the same spacetime: one time and seven space dimensions.

5. Recap: the 4+4 construction, reproduced exactly from 4+4.nb

5.1 The 8×8 matrices σ, ζ_A, τ_A, $\bar\tau_A$

With ϵ_{hpq4} the sign of the permutation $(h,p,q,4)$ (Mathematica `Signature[{h, p, q, 4}]`), 4+4.nb defines the self-dual and anti-self-dual antisymmetric 4×4 matrices

$$(s_h)_{pq} = \epsilon_{hpq4} - \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big), \qquad (t_h)_{pq} = \epsilon_{hpq4} + \big(\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}\big), \qquad h = 1,2,3,\ p,q = 1,\dots,4,$$

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},\ s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},\ s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},\ t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},\ t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

and from them the 8×8 matrices

$$\sigma = \begin{pmatrix} 0 & 1_4 \\ 1_4 & 0 \end{pmatrix}, \qquad \zeta_0 = 1_8, \qquad \zeta_h = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \qquad \zeta_{7-h} = \begin{pmatrix} 0 & t_h \\ -t_h & 0 \end{pmatrix}\ (h = 1,2,3),$$

$$\zeta_7 = \zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6 = \mathrm{diag}(-1_4, 1_4), \qquad \tau_A = \zeta_A, \qquad \bar\tau_0 = 1_8, \quad \bar\tau_A = \sigma\,\tau_A^{\mathsf{T}}\sigma\ (A = 1,\dots,7).$$

$$\tau_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_3 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_4 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix},$$

$$\tau_5 = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_6 = \begin{pmatrix} 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_7 = \begin{pmatrix} -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1 \end{pmatrix}.$$

They satisfy (all verified in 7+1.nb exactly as in 4+4.nb)

$$\tfrac{1}{2}\big(\zeta_A\zeta_B+\zeta_B\zeta_A\big) = -\eta^{(4,4)}_{AB}\,1_8 \quad (A,B=1,\dots,7), \qquad -\eta^{(4,4)}_{AA}\,\zeta_A = \zeta_A^{\mathsf T},$$
$$\tfrac{1}{2}\big(\tau_A\bar\tau_B+\tau_B\bar\tau_A\big) = \tfrac{1}{2}\big(\bar\tau_A\tau_B+\bar\tau_B\tau_A\big) = \eta^{(4,4)}_{AB}\,1_8 \quad (A,B=0,\dots,7),$$
$$\tau_1\tau_2\cdots\tau_7 = 1_8, \qquad \sigma=\tau_1\tau_2\tau_3=\tau_4\tau_5\tau_6\tau_7, \qquad \sigma^2=1_8,$$
$$\sigma\tau_h=\tau_h\sigma\ (h=1,2,3), \qquad \sigma\tau_k=-\tau_k\sigma\ (k=4,\dots,7).$$

So τ_1,τ_2,τ_3 are antisymmetric and square to -1_8; $\tau_4,\dots,\tau_7$ are symmetric and square to $+1_8$: they are the left multiplications of the three imaginary and the four split units of the split octonions.

5.2 The 16×16 Dirac matrices T_A of 4+4.nb

$$T_A = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}\ (A=0,\dots,7), \qquad T_AT_B+T_BT_A = 2\,\eta^{(4,4)}_{AB}\,1_{16},$$
$$T_8 = T_0T_1\cdots T_7 = \mathrm{diag}(-1_8,1_8), \qquad \sigma_{16} = T_0T_1T_2T_3 = \mathrm{diag}(-\sigma,\sigma),$$
$$P_L = \tfrac{1}{2}(1_{16}-T_8) = \mathrm{diag}(1_8,0), \qquad P_R = \tfrac{1}{2}(1_{16}+T_8) = \mathrm{diag}(0,1_8).$$

5.3 How the split-octonion multiplication constants come from τ_A

Choose the unit spinor $u=(1,0,0,0,1,0,0,0)^{\mathsf T}/\sqrt{2}$, an eigenvector of σ: $\sigma u = u$, $u^{\mathsf T}u = u^{\mathsf T}\sigma u = 1$. The vielbein E_{44} (4+4.nb: `E^A_a[hUSE]`) and its inverse are

$$(E_{44})_{Ba} = \big(u^{\mathsf T}\sigma\,\bar\tau_B\big)_a, \qquad E_{44}^{-1} = (u^{\mathsf T}\sigma u)\,\big(\eta^{(4,4)}_{00}\tau_0 u,\ \eta^{(4,4)}_{11}\tau_1 u,\ \dots,\ \eta^{(4,4)}_{77}\tau_7 u\big) \quad \text{(columns)}.$$

The left multiplication by the split unit ε_B (4+4.nb: `eA[B]`, $B=1,\dots,8$) is the matrix $g_{BB}\,E_{44}\,\tau_{B-1}\,E_{44}^{-1}$ with $g=\eta^{(4,4)}$:

$$\varepsilon_B\,\varepsilon_C = \sum_{D=1}^{8}\Big(g_{BB}\,E_{44}\,\tau_{B-1}\,E_{44}^{-1}\Big)_{DC}\,\varepsilon_D.$$

In words: the split-octonion multiplication constants are the matrix elements of the 8×8 matrices τ_A, written in the octonion basis by the vielbein, times the metric sign g_{BB}. This is the table `gridx` of 4+4.nb (row ε_B times column ε_C):

·	ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7	ε_8
ε_1	ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7	ε_8
ε_2	ε_2	$-\varepsilon_1$	$-\varepsilon_4$	ε_3	$-\varepsilon_6$	ε_5	ε_8	$-\varepsilon_7$
ε_3	ε_3	ε_4	$-\varepsilon_1$	$-\varepsilon_2$	$-\varepsilon_7$	$-\varepsilon_8$	ε_5	ε_6
ε_4	ε_4	$-\varepsilon_3$	ε_2	$-\varepsilon_1$	$-\varepsilon_8$	ε_7	$-\varepsilon_6$	ε_5
ε_5	ε_5	ε_6	ε_7	ε_8	ε_1	ε_2	ε_3	ε_4
ε_6	ε_6	$-\varepsilon_5$	ε_8	$-\varepsilon_7$	$-\varepsilon_2$	ε_1	$-\varepsilon_4$	ε_3
ε_7	ε_7	$-\varepsilon_8$	$-\varepsilon_5$	ε_6	$-\varepsilon_3$	ε_4	ε_1	$-\varepsilon_2$
ε_8	ε_8	ε_7	$-\varepsilon_6$	$-\varepsilon_5$	$-\varepsilon_4$	$-\varepsilon_3$	ε_2	ε_1

Split octonions: $\varepsilon_1 = 1$, $\varepsilon_2^2 = \varepsilon_3^2 = \varepsilon_4^2 = -1$, $\varepsilon_5^2 = \cdots = \varepsilon_8^2 = +1$, norm $N_S(x) = x^{\mathsf{T}}\eta^{(4,4)}x$ with $N_S(xy) = N_S(x)N_S(y)$, alternative, and with zero divisors: $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$.

6. The octonions: a real realization mirroring the split octonions

6.1 From τ to τ^O

All seven imaginary octonion units must square to -1, but $\tau_4, \ldots, \tau_7$ square to $+1$. Because σ commutes with τ_1, τ_2, τ_3, anticommutes with $\tau_4, \ldots, \tau_7$ and squares to 1_8,

$$(\tau_k\sigma)(\tau_k\sigma) = -\tau_k\tau_k\,\sigma\sigma = -1_8 \qquad (k = 4, \ldots, 7),$$

and all anticommutation relations survive. Over $\mathbb{C}$ one would multiply τ_k by i; the real matrix σ replaces that i. Definition:

$$\tau_0^O = 1_8, \qquad \tau_A^O = \tau_A \ (A = 1,2,3), \qquad \tau_A^O = \tau_A\,\sigma \ (A = 4, \ldots, 7), \qquad \bar\tau_A^O = g^O(\tau_A^O)^{\mathsf{T}}g^O = (\tau_A^O)^{\mathsf{T}}.$$

$$\tau_4^O = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}, \quad \tau_5^O = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \end{pmatrix},$$

$$\tau_6^O = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_7^O = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

and $\tau_1^O = \tau_1$, $\tau_2^O = \tau_2$, $\tau_3^O = \tau_3$ as displayed in section 5.1. Properties (verified):

$$(\tau_i^O)^{\mathsf{T}} = -\tau_i^O, \qquad (\tau_i^O)^2 = -1_8, \qquad \tau_i^O\tau_j^O = -\tau_j^O\tau_i^O \ (i \neq j),$$
$$\tau_A^O\bar\tau_B^O + \tau_B^O\bar\tau_A^O = \bar\tau_A^O\tau_B^O + \bar\tau_B^O\tau_A^O = 2\,g_{AB}^O\,1_8 = 2\,\delta_{AB}\,1_8,$$
$$\tau_1^O\tau_2^O\cdots\tau_7^O = 1_8 \qquad (\text{as } \tau_1\cdots\tau_7 = 1_8 \text{ in 4+4.nb}),$$
$$\tau_4^O = \tau_5\tau_6\tau_7, \quad \tau_5^O = -\tau_4\tau_6\tau_7, \quad \tau_6^O = \tau_4\tau_5\tau_7, \quad \tau_7^O = -\tau_4\tau_5\tau_6.$$

For $x = \sum_A x_A e_A$ put $\tau^O(x) = \sum_A x_A\tau_A^O$; then $\tau^O(x)\,\bar\tau^O(x) = N(x)\,1_8$ with $N(x) = x_0^2 + \cdots + x_7^2$, the matrix form of $x\bar x = N(x)$. In 4+4.nb the same relation holds with $\eta^{(4,4)}$, the split norm.

6.2 The octonion vielbein (same unit spinor u)

In 4+4.nb the rows of the vielbein are $u^{\mathsf{T}}\sigma\bar{\tau}_B$, with the split spinor metric σ. For the octonions the spinor metric is Euclidean ($\sigma \to g^O = 1_8$), so

$$(E^O)_{Ba} = \big(u^{\mathsf{T}} g^O \bar{\tau}^O_B\big)_a, \qquad (E^O)^{-1} = (E^O)^{\mathsf{T}} = (u^{\mathsf{T}} g^O u)\big(g^O_{00}\tau^O_0 u, \ldots, g^O_{77}\tau^O_7 u\big), \qquad E^O = \eta^{(4,4)} E_{44}.$$

Because $\sigma u = u$, $\tau^O_A u = \tau_A u$ for every A: the two vielbeins differ only by the signs of rows 5–8.

$$E^O = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{pmatrix}.$$

6.3 The octonion multiplication constants

Exactly the formula of 4+4.nb, with $\tau \to \tau^O$, $E_{44} \to E^O$, $g \to g^O$ and a 0-based basis $e_A \leftrightarrow \tau^O_A \leftrightarrow x_A$:

$$L^O_A = g^O_{AA}\, E^O\, \tau^O_A\, (E^O)^{-1}, \qquad e_A\, e_K = \sum_{L=0}^{7} \big(L^O_A\big)_{LK}\, e_L, \qquad (xy)_L = \sum_{A,K} x_A\, y_K\, \big(L^O_A\big)_{LK}.$$

Every L^O_A is a signed permutation matrix, so every multiplication constant is 0 or ± 1. The table (row e_A times column e_K; Mathematica `gridO`, the mirror of `gridx`):

·	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	$-e_3$	e_2	$-e_5$	e_4	e_7	$-e_6$
e_2	e_2	e_3	$-e_0$	$-e_1$	$-e_6$	$-e_7$	e_4	e_5
e_3	e_3	$-e_2$	e_1	$-e_0$	$-e_7$	e_6	$-e_5$	e_4
e_4	e_4	e_5	e_6	e_7	$-e_0$	$-e_1$	$-e_2$	$-e_3$
e_5	e_5	$-e_4$	e_7	$-e_6$	e_1	$-e_0$	e_3	$-e_2$
e_6	e_6	$-e_7$	$-e_4$	e_5	e_2	$-e_3$	$-e_0$	e_1
e_7	e_7	e_6	$-e_5$	$-e_4$	e_3	e_2	$-e_1$	$-e_0$

6.4 The properties of [1A], verified

Units. e_0 is a two-sided identity, $e_i^2 = -e_0$ and $e_i e_j = -e_j e_i$ for $i \neq j$ ($i, j = 1, \ldots, 7$).

Fano plane. Each product of two distinct imaginary units is $\pm$ a third one. The seven triples are the lines of the Fano plane; written in the cyclic order (a, b, c) with $e_a e_b = +e_c$ (and then $e_b e_c = e_a$, $e_c e_a = e_b$):

$$(1,3,2),\ (1,5,4),\ (1,6,7),\ (2,6,4),\ (2,7,5),\ (3,5,6),\ (3,7,4).$$

Every pair of distinct imaginary units lies on exactly one line, and each line together with e_0 spans a copy of the quaternions (closed and associative). In a picture: put e_1 at the centre, e_2, e_4, e_7 at the vertices of a triangle and e_6, e_3, e_5 at the midpoints of the sides e_2e_4, e_4e_7, e_7e_2; the lines are the three sides, the three medians and the inscribed circle through e_3, e_5, e_6.

Algebraic laws (proved symbolically for general octonions x, y, z with 8 symbolic components each; the Moufang identities, being linear in x and y, are proved for basis units x, y and symbolic z):

$$x(xy) = (xx)y, \qquad (yx)x = y(xx), \qquad (xy)x = x(yx), \qquad x(xx) = (xx)x,$$

$$
\begin{aligned}
[x,y,z] &:= (xy)z - x(yz) = -[y,x,z] = -[x,z,y],\\
z\big(x(zy)\big) &= \big((zx)z\big)y, \qquad x\big(z(yz)\big) = \big((xz)y\big)z, \qquad (zx)(yz) = \big(z(xy)\big)z,\\
x\bar{x} &= \bar{x}x = N(x)\,e_0, \qquad \overline{xy} = \bar{y}\,\bar{x}, \qquad \mathrm{Re}(x) = \tfrac{1}{2}(x+\bar{x}) = x_0,\\
N(xy) &= N(x)\,N(y), \qquad x\,x^{-1} = x^{-1}x = e_0 \ \text{ for } \ x^{-1} = \bar{x}/N(x),\\
x^2 &- 2\,\mathrm{Re}(x)\,x + N(x)\,e_0 = 0.
\end{aligned}
$$

The last identity shows that the subalgebra generated by one octonion is $\mathbb{R}$ or $\mathbb{C}$ (power-associativity). Multiplicativity of the norm is Degen's eight-square identity: with $x = \sum a_A e_A$ and $y = \sum b_A e_A$ the components of xy are

$$
\begin{aligned}
(xy)_0 &= a_0b_0 - a_1b_1 - a_2b_2 - a_3b_3 - a_4b_4 - a_5b_5 - a_6b_6 - a_7b_7,\\
(xy)_1 &= a_0b_1 + a_1b_0 - a_2b_3 + a_3b_2 - a_4b_5 + a_5b_4 + a_6b_7 - a_7b_6,\\
(xy)_2 &= a_0b_2 + a_1b_3 + a_2b_0 - a_3b_1 - a_4b_6 - a_5b_7 + a_6b_4 + a_7b_5,\\
(xy)_3 &= a_0b_3 - a_1b_2 + a_2b_1 + a_3b_0 - a_4b_7 + a_5b_6 - a_6b_5 + a_7b_4,\\
(xy)_4 &= a_0b_4 + a_1b_5 + a_2b_6 + a_3b_7 + a_4b_0 - a_5b_1 - a_6b_2 - a_7b_3,\\
(xy)_5 &= a_0b_5 - a_1b_4 + a_2b_7 - a_3b_6 + a_4b_1 + a_5b_0 + a_6b_3 - a_7b_2,\\
(xy)_6 &= a_0b_6 - a_1b_7 - a_2b_4 + a_3b_5 + a_4b_2 - a_5b_3 + a_6b_0 + a_7b_1,\\
(xy)_7 &= a_0b_7 + a_1b_6 - a_2b_5 - a_3b_4 + a_4b_3 + a_5b_2 - a_6b_1 + a_7b_0,
\end{aligned}
$$

$$
\Big(\sum_{A=0}^{7} a_A^2\Big)\Big(\sum_{A=0}^{7} b_A^2\Big) = \sum_{L=0}^{7} (xy)_L^2 .
$$

Hence $N(x) = 0$ only for $x = 0$, and $xy = 0$ only if $x = 0$ or $y = 0$: no zero divisors, a division algebra (unlike the split octonions).

Non-associativity. Exactly 168 of the 512 ordered basis triples (e_A, e_B, e_K) are non-associative: the triples of distinct imaginary units not on a common Fano line ($7 \cdot 6 \cdot 4 = 168$). Example: $(e_1e_2)e_4 = e_7$ but $e_1(e_2e_4) = -e_7$.

A numerical example. For $x = e_0 + 2e_1 - e_4 + 3e_7$, $y = -2e_0 + e_2 + e_5$, $z = e_3 + e_6$:

$$
\begin{aligned}
xy &= -2e_0 - 3e_1 + 4e_2 - 2e_3 + 4e_4 - 2e_5 - e_6 - 6e_7,\\
yx &= -2e_0 - 5e_1 - 2e_2 + 2e_3 + 4e_5 + e_6 - 6e_7,\\
N(x) &= 15, \quad N(y) = 6, \quad N(xy) = 90 = N(x)N(y),\\
x^{-1} &= \frac{1}{15}e_0 - \frac{2}{15}e_1 + \frac{1}{15}e_4 - \frac{1}{5}e_7,\\
(xy)z - x(yz) &= -6e_1 - 2e_2 - 6e_3 + 6e_4 + 2e_5 + 6e_6 + 6e_7 \neq 0.
\end{aligned}
$$

6.5 Cayley–Dickson doubling $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

The span of e_0, e_1, e_2, e_3 — the time direction and the three ordinary space directions $x_0, \ldots, x_3$ — is a quaternion subalgebra $\mathbb{H}$, and $\ell = e_4$ doubles it: $e_4e_k = e_{4+k}$ $(k = 1,2,3)$, $e_4^2 = -e_0$. Every octonion is $x = a + \ell b$ with quaternions $a = (x_0, x_1, x_2, x_3)$ and $b = (x_4, x_5, x_6, x_7)$. The orientation inherited from 4+4.nb is $e_1e_2 = -e_3$ (so $i = e_1$, $j = e_2$, $k = -e_3$ are Hamilton's units). With $\ell^2 = \gamma$ the product is

$$
\begin{aligned}
(a,b)(c,d) &= \big(ac + \gamma\, d\bar{b},\ \ \bar{a}d + cb\big),\\
\gamma = -1 &: \ \text{octonions (this document)}, \qquad \gamma = +1 : \ \text{split octonions (4+4.nb)}.
\end{aligned}
$$

Both cases are proved symbolically in 7+1.nb. The formula $(ac - \bar{d}b,\ \bar{a}d + cb)$ of [1A] fails: for $x = (1,1)$, $y = (i,0)$ it gives $x^2y = (0, 2i)$ but $x(xy) = (2i, 2i)$, and its norm is not multiplicative.

6.6 Octonions versus split octonions

With the split units relabelled 0-based ($\varepsilon_{B+1} \to e^S_B$), the two tables differ only in the products of two doubled units:

$$e_A e_B\big|_{\mathbb{O}} = s_{AB}\; e^S_A e^S_B\big|_{\text{split}}, \qquad s_{AB} = \begin{cases} -1, & A \geq 4 \text{ and } B \geq 4, \\ +1, & \text{otherwise,} \end{cases}$$

because $\ell^2 = -1$ instead of $+1$. Over $\mathbb{C}$ the algebras coincide: $\varphi(e^S_A) = c_A e_A$ with $c = (1,1,1,1,i,i,i,i)$ satisfies $\varphi(xy) = \varphi(x)\varphi(y)$ and $N_{\mathbb{O}}(\varphi(x)) = N_S(x)$, so $\mathbb{O}_{\text{split}} \otimes \mathbb{C} \cong \mathbb{O} \otimes \mathbb{C}$. The real substitute for the factors i is the matrix σ of section 6.1.

7. Clifford algebras from the octonions

7.1 The exact mirror of T_A: the Euclidean algebra $Cl(8,0)$

Copying $T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$ with $\tau \to \tau^O$:

$$T^O_A = \begin{pmatrix} 0 & \bar\tau^O_A \\ \tau^O_A & 0 \end{pmatrix}, \qquad T^O_A T^O_B + T^O_B T^O_A = 2\, g^O_{AB}\, 1_{16} = 2\,\delta_{AB}\, 1_{16} \qquad (Cl(8,0) \cong \mathbb{R}(16)),$$
$$T^O_A = T_A \;(A = 0,1,2,3), \qquad T^O_k = T_k\, S_{44} \;(k = 4,\dots,7), \qquad S_{44} = T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma,\sigma),$$
$$T^O_8 = T^O_0 T^O_1 \cdots T^O_7 = T_8 = \mathrm{diag}(-1_8, 1_8).$$

7.2 The real Clifford algebra of 7+1 spacetime: $\Gamma^{(1,7)}$, $Cl(1,7) \cong \mathbb{R}(16)$

Definition.

$$\Gamma^{(1,7)}_0 = T^O_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix} \;\text{(time)}, \qquad \Gamma^{(1,7)}_i = T^O_0\, T^O_i = \begin{pmatrix} \tau^O_i & 0 \\ 0 & \bar\tau^O_i \end{pmatrix} \;(i = 1,\dots,7,\ \text{space}),$$
$$\Gamma^{(1,7)}_0 = T_0, \qquad \Gamma^{(1,7)}_i = T_0\, T_i \;(i = 1,2,3), \qquad \Gamma^{(1,7)}_k = T_0\, T_k\, S_{44} \;(k = 4,\dots,7) \qquad \text{(only 4+4.nb matrices)}.$$

Relations (all verified):

$$\Gamma^{(1,7)}_A \Gamma^{(1,7)}_B + \Gamma^{(1,7)}_B \Gamma^{(1,7)}_A = 2\,\eta^{(1,7)}_{AB}\, 1_{16} = -2\,\eta^{(7,1)}_{AB}\, 1_{16},$$
$$(\Gamma^{(1,7)}_0)^{\mathsf T} = \Gamma^{(1,7)}_0, \quad (\Gamma^{(1,7)}_i)^{\mathsf T} = -\Gamma^{(1,7)}_i,$$
$$(\Gamma^{(1,7)}_A)^{\mathsf T} = \Gamma^{(1,7)}_0\, \Gamma^{(1,7)}_A\, \Gamma^{(1,7)}_0 \quad \text{(real Dirac adjoint)},$$
$$\mathrm{tr}\big(\Gamma^{(1,7)}_{A_1} \cdots \Gamma^{(1,7)}_{A_k}\big) = 0 \quad (1 \leq k \leq 8,\ A_1 < \cdots < A_k).$$

All entries are 0 or ± 1. **Why this is the Clifford algebra of [1B].** (i) For $v = \sum_A v_A e_A$ the matrix $\Gamma^{(1,7)}(v) = \sum_A v_A \Gamma^{(1,7)}_A$ satisfies

$$\Gamma^{(1,7)}(v)^2 = Q(v)\, 1_{16}, \qquad Q(v) = v^{\mathsf T} \eta^{(1,7)} v = v_0^2 - v_1^2 - \cdots - v_7^2,$$

so by the universal property of $Cl(V,Q) = \mathcal{T}(V)/\mathcal{I}_Q(V)$ the map extends to an algebra homomorphism $Cl(V,Q) \to \mathbb{R}(16)$. (ii) The 256 ordered products $\Gamma^{(1,7)}_{A_1} \cdots \Gamma^{(1,7)}_{A_k}$ ($A_1 < \cdots < A_k$, $k = 0,\dots,8$, counted by $\binom{8}{k}$) are linearly independent, so the homomorphism is injective; both sides have dimension 256, hence $Cl(1,7) \cong \mathbb{R}(16)$. (iii) Only multiples of 1_{16} commute with all $\Gamma^{(1,7)}_A$: the representation is irreducible over $\mathbb{R}$.

7.3 The complexified realization in the exact block form of 4+4.nb: $\Gamma^{(7,1)}$

For the sign convention of 4+4.nb, $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta^{(7,1)}_{AB}1_{16}$, complex numbers are unavoidable (section 7.4). The complexification needed is minimal: multiply the time block by i.

$$\tau_0^C = \bar\tau_0^C = i\,1_8, \qquad \tau_k^C = \tau_k^O, \quad \bar\tau_k^C = \bar\tau_k^O \ (k=1,\dots,7), \qquad \Gamma^{(7,1)}_A = \begin{pmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{pmatrix},$$

$$\Gamma^{(7,1)}_0 = i\,T_0, \qquad \Gamma^{(7,1)}_k = T_k^O \ (k=1,\dots,7), \qquad \Gamma^{(7,1)}_A\Gamma^{(7,1)}_B + \Gamma^{(7,1)}_B\Gamma^{(7,1)}_A = 2\,\eta^{(7,1)}_{AB}\,1_{16},$$

$$\Gamma^{(7,1)}_A = S\left(i\,\Gamma^{(1,7)}_A\right)S^{-1}, \qquad S = \tfrac{1}{\sqrt2}\big(1_{16} - i\,\Gamma^{(1,7)}_0\big), \qquad SS^\dagger = 1_{16}.$$

This is literally the formula $T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$ of 4+4.nb, applied to the complexified 8×8 matrices; only the time component is complex.

7.4 Theorem: with $\eta^{(7,1)}$ there is no real 16×16 realization

Theorem. There are no real 16×16 matrices $R_0,\dots,R_7$ with $R_AR_B + R_BR_A = 2\eta^{(7,1)}_{AB}1_{16}$. Reordering, renaming or changing the signs of the indices cannot help, because none of these changes the signature $(7,1)$.

Proof. $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has exactly one irreducible module, $\mathbb{C}^{16}$. If real R_A existed, R_A and $\Gamma^{(7,1)}_A$ would be two 16-dimensional modules, so $R_A = T\,\Gamma^{(7,1)}_A\,T^{-1}$ for some invertible complex T. Since R_A is real, complex conjugation gives

$$M\,\overline{\Gamma^{(7,1)}_A} = \Gamma^{(7,1)}_A\,M \quad (A=0,\dots,7), \qquad M = T^{-1}\,\overline{T}, \qquad M\,\overline{M} = 1_{16}.$$

By Schur's lemma all such M form a one-dimensional space. Solving the linear equations (7.4 of the notebook) gives exactly that space, spanned by the volume element

$$M_0 = \Gamma^{(1,7)}_8 = \Gamma^{(1,7)}_0\Gamma^{(1,7)}_1\cdots\Gamma^{(1,7)}_7 = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad M_0\,\overline{M_0} = M_0^2 = -1_{16}.$$

Hence $M = cM_0$ and $M\overline{M} = -|c|^2\,1_{16} \neq 1_{16}$, a contradiction. □ The sign of $M\overline{M}$ is a Frobenius–Schur-type indicator: $+1$ means the complex module has a real form (the algebra is $\mathbb{R}(16)$), -1 means it is quaternionic (the algebra is $\mathbb{H}(8)$, whose smallest real module is $\mathbb{H}^8 = \mathbb{R}^{32}$). For $\Gamma^{(1,7)}$ the indicator is $+1$ (take $M = 1_{16}$).

7.5 A real realization of $Cl(7,1)$ needs 32×32 matrices

$$\widehat\Gamma^{(7,1)}_A = \varepsilon\otimes\Gamma^{(1,7)}_A, \qquad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \widehat\Gamma^{(7,1)}_A\widehat\Gamma^{(7,1)}_B + \widehat\Gamma^{(7,1)}_B\widehat\Gamma^{(7,1)}_A = 2\,\eta^{(7,1)}_{AB}\,1_{32},$$

$$Q_i = \varepsilon\otimes 1_{16}, \qquad Q_j = \sigma_1\otimes\Gamma^{(1,7)}_8, \qquad Q_k = \sigma_3\otimes\Gamma^{(1,7)}_8,$$

$$Q_i^2 = Q_j^2 = Q_k^2 = -1_{32}, \qquad Q_iQ_j = Q_k = -Q_jQ_i, \qquad Q\,\widehat\Gamma^{(7,1)}_A = \widehat\Gamma^{(7,1)}_A\,Q \quad (Q = Q_i,Q_j,Q_k).$$

The matrices commuting with all $\widehat\Gamma^{(7,1)}_A$ form exactly the 4-dimensional span of $1_{32}, Q_i, Q_j, Q_k$ (computed in the notebook): the quaternions. This is $Cl(7,1) \cong \mathbb{H}(8)$ acting on $\mathbb{R}^{32} = \mathbb{H}^8$.

7.6 Volume elements and chirality

$$T_8^2 = +1_{16} \ \Rightarrow\ P_{L,R} = \tfrac12(1_{16} \mp T_8) \text{ real (4+4.nb)}; \qquad (T_8^O)^2 = +1_{16},\ T_8^O = T_8,$$

$$\big(\Gamma^{(1,7)}_8\big)^2 = -1_{16}, \qquad \Gamma^{(1,7)}_8\Gamma^{(1,7)}_A = -\Gamma^{(1,7)}_A\Gamma^{(1,7)}_8, \qquad \Gamma^{(1,7)}_8\Sigma_{AB} = \Sigma_{AB}\Gamma^{(1,7)}_8,$$

$$\chi^{(7,1)} = -i\,\Gamma^{(7,1)}_8 = T_8^O = T_8, \qquad \tfrac12\big(1_{16} \mp \chi^{(7,1)}\big) = P_{L,R},$$

$$\chi^{(1,7)} = i\,\Gamma^{(1,7)}_8, \qquad \tfrac12\big(1_{16} \mp \chi^{(1,7)}\big) \text{ complex projectors of rank 8 each.}$$

With real matrices in 7+1 dimensions there is no real chirality projector: $\Gamma^{(1,7)}_8$ is a complex structure commuting with Spin(1,7), so the real 16-component spinor is one complex 8-component Weyl spinor. In the basis of $\Gamma^{(7,1)}$ the chiral projectors are literally P_L and P_R of 4+4.nb.

7.7 What the octonions do: $\Gamma(x)^2 = \mathrm{Re}(xx)$ and alternativity

$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = x^{\mathsf{T}}\eta^{(1,7)}x, \qquad \Gamma^{(1,7)}(x)^2 = \mathrm{Re}(x\,x)\,1_{16},$$
$$T^O(x)^2 = N(x)\,1_{16} = x^{\mathsf{T}}g^O x\,1_{16}, \qquad T(x)^2 = x^{\mathsf{T}}\eta^{(4,4)}x\,1_{16}\ \text{(4+4.nb)}.$$

Rotating both 8-blocks by E^O gives $\Gamma^{(1,7)}$ in the octonion basis,

$$\Gamma^e_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma^e_i = \begin{pmatrix} L^O_i & 0 \\ 0 & -L^O_i \end{pmatrix}, \qquad \Gamma^e(x)\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \begin{pmatrix} x_0\psi_2 + \vec{x}\,\psi_1 \\ x_0\psi_1 - \vec{x}\,\psi_2\end{pmatrix} \quad (\psi_1,\psi_2 \in \mathbb{O}),$$

$$\Gamma^e(x)^2\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = \begin{pmatrix} x_0^2\psi_1 + \vec{x}(\vec{x}\psi_1) \\ x_0^2\psi_2 + \vec{x}(\vec{x}\psi_2)\end{pmatrix} = (x_0^2 - |\vec{x}|^2)\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix},$$

where $\vec{x} = \sum_i x_i e_i$ and the last step is alternativity, $\vec{x}(\vec{x}\psi) = (\vec{x}\vec{x})\psi = -|\vec{x}|^2\psi$. The Clifford relation of 7+1 spacetime is the alternativity of the octonions.

8. Examples that test everything

8.1 The Lorentz algebra so(1,7)

$$\Sigma_{AB} = \tfrac{1}{4}\big(\Gamma^{(1,7)}_A\Gamma^{(1,7)}_B - \Gamma^{(1,7)}_B\Gamma^{(1,7)}_A\big) \quad (A<B:\ 28\ \text{real, linearly independent matrices}),$$
$$\big[\Sigma_{AB}, \Gamma^{(1,7)}_K\big] = \eta^{(1,7)}_{BK}\,\Gamma^{(1,7)}_A - \eta^{(1,7)}_{AK}\,\Gamma^{(1,7)}_B,$$
$$\big[\Sigma_{AB}, \Sigma_{KL}\big] = \eta^{(1,7)}_{BK}\Sigma_{AL} - \eta^{(1,7)}_{AK}\Sigma_{BL} - \eta^{(1,7)}_{BL}\Sigma_{AK} + \eta^{(1,7)}_{AL}\Sigma_{BK}.$$

8.2 A boost along x_1 and a rotation in the x_4–x_5 plane

Because $(\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1)^2 = +1_{16}$ and $(\Gamma^{(1,7)}_4\Gamma^{(1,7)}_5)^2 = -1_{16}$,

$$S(\beta) = e^{\beta\Sigma_{01}} = \cosh\tfrac{\beta}{2}\,1_{16} + \sinh\tfrac{\beta}{2}\,\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1, \qquad S(\beta)^{-1} = \cosh\tfrac{\beta}{2}\,1_{16} - \sinh\tfrac{\beta}{2}\,\Gamma^{(1,7)}_0\Gamma^{(1,7)}_1,$$
$$R(\theta) = e^{\theta\Sigma_{45}} = \cos\tfrac{\theta}{2}\,1_{16} + \sin\tfrac{\theta}{2}\,\Gamma^{(1,7)}_4\Gamma^{(1,7)}_5, \qquad R(\theta)^{-1} = \cos\tfrac{\theta}{2}\,1_{16} - \sin\tfrac{\theta}{2}\,\Gamma^{(1,7)}_4\Gamma^{(1,7)}_5,$$

both real (they solve $dS/d\beta = \Sigma_{01}S$, $S(0) = 1_{16}$ and $dR/d\theta = \Sigma_{45}R$, $R(0) = 1_{16}$). They act on the generators as Lorentz transformations, with Λ read off from traces:

$$S\,\Gamma^{(1,7)}_K\,S^{-1} = \sum_{B=0}^{7}\Lambda_{BK}\,\Gamma^{(1,7)}_B, \qquad \Lambda_{BK} = \tfrac{1}{16}\,\mathrm{tr}\big(\eta^{(1,7)}_{BB}\,\Gamma^{(1,7)}_B\,S\,\Gamma^{(1,7)}_K\,S^{-1}\big), \qquad \Lambda^{\mathsf{T}}\eta^{(1,7)}\Lambda = \eta^{(1,7)},$$

$$\Lambda_{\text{boost}} = \begin{pmatrix} \cosh\beta & -\sinh\beta & 0 \\ -\sinh\beta & \cosh\beta & 0 \\ 0 & 0 & 1_6 \end{pmatrix}, \qquad \Lambda_{\text{rot}} = \begin{pmatrix} 1_4 & 0 & 0 \\ 0 & \begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix} & 0 \\ 0 & 0 & 1_2 \end{pmatrix}.$$

8.3 The 7+1 Dirac operator squares to the wave operator

$$\Gamma^A = \eta^{AA}_{(1,7)}\,\Gamma^{(1,7)}_A, \qquad \big(\Gamma^A\partial_A\big)^2\Psi = \eta^{AB}_{(1,7)}\,\partial_A\partial_B\Psi = \big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2\big)\Psi$$

for 16 completely arbitrary functions $\Psi = (\Psi_1, \ldots, \Psi_{16})$ of $x_0, \ldots, x_7$ (the notebook proves it symbolically; here $\eta^{AB}_{(1,7)}$ is the inverse metric, equal to $\eta^{(1,7)}_{AB}$).

8.4 Spinors on the mass shell: real amplitudes

Take $p = (3,2,2,0,0,0,0,0)$ (contravariant components p^A) and $m = 1$: $p^{\mathsf T}\eta^{(1,7)}p = 9-4-4 = 1 = m^2$. With $\not p = \sum_A p^A \Gamma^{(1,7)}_A$ we have $\not p^{\,2} = m^2\, 1_{16}$, so $\not p$ has eigenvalues $\pm m$, eight times each. Because $\not p$ is a real matrix, $(\not p - m)w = 0$ has an 8-dimensional space of REAL solutions w (spinor amplitudes). The basis printed by Mathematica (`NullSpace`, one solution per row):

$$W = \begin{pmatrix} 2 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ -2 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & -2 & -2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 2 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 2 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -2 & -2 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad (\not p - m)\, W^{\mathsf T} = 0.$$

Each row w gives the plane wave $\Psi = w\, e^{-i\,p\cdot x}$, $p\cdot x = p^{\mathsf T}\eta^{(1,7)}x$, solving $(i\Gamma^A\partial_A - m)\Psi = 0$. The plane wave itself is complex: real amplitudes do not make a real field (section 8.5).

8.5 Real spinor fields: massless with 16 components, massive with 32

The operator $i\Gamma^A\partial_A$ is imaginary, so a real Ψ cannot solve $(i\Gamma^A\partial_A - m)\Psi = 0$. A real first-order equation $(G^A\partial_A - m)\Psi = 0$ with real matrices G_A ($G^A = \eta^{AA}_{(7,1)}G_A$) implies the Klein–Gordon equation, through

$$\big(G^A\partial_A + m\big)\big(G^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big),$$

only if $G_AG_B + G_BG_A = 2\,\eta^{(7,1)}_{AB}\,1$ (time squaring to -1), and that relation has no real 16×16 solution (section 7.4). The remaining real option, $(\Gamma^A\partial_A - mJ)\Psi = 0$, squares to the Klein–Gordon operator $\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2$ only if the real matrix J anticommutes with every $\Gamma^{(1,7)}_A$ and $J^2 = +1_{16}$. But (computed in the notebook)

$$\big\{J \in \mathbb{R}(16):\ J\,\Gamma^{(1,7)}_A = -\Gamma^{(1,7)}_A\,J\ \ (A = 0,\ldots,7)\big\} = \mathbb{R}\,\Gamma^{(1,7)}_8, \qquad \big(c\,\Gamma^{(1,7)}_8\big)^2 = -c^2\,1_{16},$$

and with $J = c\,\Gamma^{(1,7)}_8$ the square is $\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 - c^2m^2$, a tachyon. So with 16 real components only massless real fields exist. For the null momentum $k = (1,1,0,0,0,0,0,0)$, $k^{\mathsf T}\eta^{(1,7)}k = 0$ and $\not k = \sum_A k^A\Gamma^{(1,7)}_A$ satisfies $\not k^{\,2} = 0$; the real equation $\not k\, w = 0$ has 8 real solutions, and for each of them

$$\Psi = w\cos(k\cdot x), \qquad \Gamma^A\partial_A\Psi = -\sin(k\cdot x)\ \not k\, w = 0 \qquad \text{(a real solution)}.$$

A real massive field needs the real 32×32 matrices $\widehat\Gamma^{(7,1)}_A = \varepsilon\otimes\Gamma^{(1,7)}_A$ of section 7.5, which do satisfy the $\eta^{(7,1)}$ relation. With $\widehat\Gamma^A = \eta^{AA}_{(7,1)}\widehat\Gamma^{(7,1)}_A$ and $P = \sum_A p^A\widehat\Gamma^{(7,1)}_A$ (p as in section 8.4):

$$\big(\widehat\Gamma^A\partial_A\big)^2 = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2\big)\,1_{32}, \qquad P^2 = \big(p^{\mathsf T}\eta^{(7,1)}p\big)\,1_{32} = -m^2\,1_{32},$$
$$\Psi = w\cos(p\cdot x) + \tfrac{1}{m}\,P\,w\,\sin(p\cdot x) \qquad (w \in \mathbb{R}^{32}\ \text{arbitrary}),$$
$$\big(\widehat\Gamma^A\partial_A - m\big)\Psi = 0, \qquad \big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)\Psi = 0.$$

Derivation: with $\phi = p\cdot x$ one has $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta^{AA}_{(7,1)}\eta^{(1,7)}_{AA} = -1$, so $\widehat\Gamma^A\partial_A\Psi = -P\big(-w\sin\phi + \frac{1}{m}Pw\cos\phi\big) = Pw\sin\phi + m\,w\cos\phi = m\Psi$. The 32 real components are the quaternionic module $\mathbb{H}^8 = \mathbb{R}^{32}$ of $Cl(7,1) \cong \mathbb{H}(8)$.

8.6 All nine signatures $p + q = 8$

Complex generators of every signature: the first p of the T_A^O and i times the remaining q. The indicator of section 7.4 then decides $\mathbb{R}(16)$ or $\mathbb{H}(8)$ (table of section 4.3). Real 16×16 generators exist exactly in the five $\mathbb{R}(16)$ cases, and all five are built here from the 4+4 and octonion matrices:

$$
\begin{aligned}
&Cl(8,0):\ T_A^O, \qquad Cl(4,4):\ T_A\ \text{(4+4.nb)}, \qquad Cl(1,7):\ \Gamma_A^{(1,7)},\\
&Cl(5,3):\ \Gamma_0^{(5,3)} = T_0,\ \ \Gamma_i^{(5,3)} = T_0 T_i\ (i = 1,\dots,7), \qquad \eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1),\\
&\qquad \Gamma_A^{(5,3)}\Gamma_B^{(5,3)} + \Gamma_B^{(5,3)}\Gamma_A^{(5,3)} = 2\,\eta_{AB}^{(5,3)} 1_{16},\\
&Cl(0,8):\ \Gamma_A^{(0,8)} = T_8^O\, T_A^O, \qquad \Gamma_A^{(0,8)}\Gamma_B^{(0,8)} + \Gamma_B^{(0,8)}\Gamma_A^{(0,8)} = -2\,\delta_{AB} 1_{16}.
\end{aligned}
$$

$\Gamma^{(5,3)}$ is the split-octonion analogue of $\Gamma^{(1,7)}$: its quadratic form $x^{\mathsf{T}}\eta^{(5,3)}x$ is $\mathrm{Re}(xx)$ of the split octonions, just as $x^{\mathsf{T}}\eta^{(1,7)}x$ is $\mathrm{Re}(xx)$ of the octonions.

9. Verification

In Mathematica. 7+1.nb was evaluated in 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026) with `NotebookEvaluate` and `InsertResults -> True`, in the private context `Notebook$$43$40478`. It contains 56 input cells and 128 checks of the form `check["name", test]`; all 128 returned True, the summary cells printed `failedChecks = {}` and `allChecksPassed = True`, there were 0 message cells, and the evaluation took 5.09545 s.

notebook section	checks	True
2. Recap: the 4+4 objects, reproduced exactly from 4+4.nb	24	24
3. The octonions: a real realization mirroring the split octonions	47	47
4. Clifford algebras from the octonions: 16×16 matrices	32	32
5. Examples that test everything	25	25

Independently in Python (exact arithmetic with sympy, in `tools/`): `ref44.py` rebuilds every 4+4.nb object and reproduces the split table of 4+4.nb; `ref71.py` and `extra71.py` rebuild every 7+1 object and re-prove the relations; `verify_7p1_nb.py` parses the evaluated 7+1.nb and confirms that the matrices τ_A, τ_A^O, $\Gamma_A^{(1,7)}$, the two multiplication tables, the matrices Λ, the eight real amplitudes W and the classification table printed by Mathematica coincide with the Python values.

10. The notebook 7+1.nb

- **0. How to use this notebook** — evaluation, the `check` bookkeeping, the private context, index conventions.
- **1. Conventions, coordinates and notation** — the coordinate table, the metrics $\eta^{(4,4)}$, $\eta^{(7,1)}$, $\eta^{(1,7)}$, g^O, the dictionary of names, the definitions [1A], [1B] and the four corrections.
- **2. Recap of 4+4.nb** — σ, s_h, t_h, ζ_A, τ_A, $\bar\tau_A$, T_A, P_L, P_R, u, E_{44}, the split table `gridx44` compared with the table printed by 4+4.nb, split product, zero divisors.
- **3. The octonions** — τ^O, vielbein E^O, `grid0`, Fano plane (with a picture), the laws of [1A], the eight-square identity, non-associativity, a numerical example, Cayley–Dickson, comparison with the split octonions.
- **4. Clifford algebras** — T^O ($Cl(8,0)$), $\Gamma^{(1,7)}$ (real 7+1), $\Gamma^{(7,1)}$ (complexified 7+1), the theorem, $\widehat{\Gamma}^{(7,1)}$ (real 32×32) and its quaternionic commutant, chirality, the octonionic meaning.
- **5. Examples** — so(1,7), boost and rotation, Dirac operator, real on-shell spinor amplitudes, real massless (16 components) and massive (32 components) spinor fields, all nine signatures.

- **6. Summary of all checks** — `Length[checks]`, `failedChecks`, `allChecksPassed` and a table of all checks.

To re-evaluate: open 7+1.nb in Wolfram 15.0.1 and choose Evaluation ▷ Evaluate Notebook (about five seconds), or run `Get["D:/Developer/github/7+1/tools/run_eval.wl"]` from any notebook, which evaluates 7+1.nb in place and saves it. The notebook is generated by `tools/make_7p1_nb.py` (`python3 tools/make_7p1_nb.py` writes the unevaluated 7+1.nb).

11. Files in this repository

path	content
`7+1.nb`	the new notebook, evaluated, with all outputs
`4+4.nb`	the 4+4 notebook (split octonions, T_A), the starting point; unchanged
`NASH_Mathematica_Notebooks/`	earlier notebooks of the project; unchanged
`tools/make_7p1_nb.py`, `tools/nbwrite.py`	generator of 7+1.nb
`tools/run_eval.wl`	evaluates 7+1.nb in place in Wolfram 15.0.1
`tools/ref44.py`, `tools/ref71.py`, `tools/extra71.py`, `tools/octalg.py`	independent Python (sympy) implementation and checks
`tools/classify_clifford.py`	numerical classification of the nine signatures via commutants of realified representations
`tools/cayley_dickson_formulas.py`, `tools/cayley_dickson_match.py`	tests of Cayley–Dickson formulas against the tables
`tools/nbparse.py`, `tools/verify_7p1_nb.py`	parser for .nb files; cross-check of the evaluated 7+1.nb
`tools/docgen.py`, `tools/doc_*.py`, `tools/make_docs.py`	generators of the documentation (Markdown, LaTeX, PDF)

12. References

- J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205, arXiv:math/0105155 — section 2.2 gives the Cayley–Dickson product $(a,b)(c,d) = (ac - d\bar{b},\ \bar{a}d + cb)$ used in section 6.5.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press (1989) — classification of real Clifford algebras and the periodicity $Cl(p+1,q+1) \cong Cl(p,q) \otimes \mathbb{R}(2)$.
- F. Reese Harvey, *Spinors and Calibrations*, Academic Press (1990) — octonions, triality and spinors.